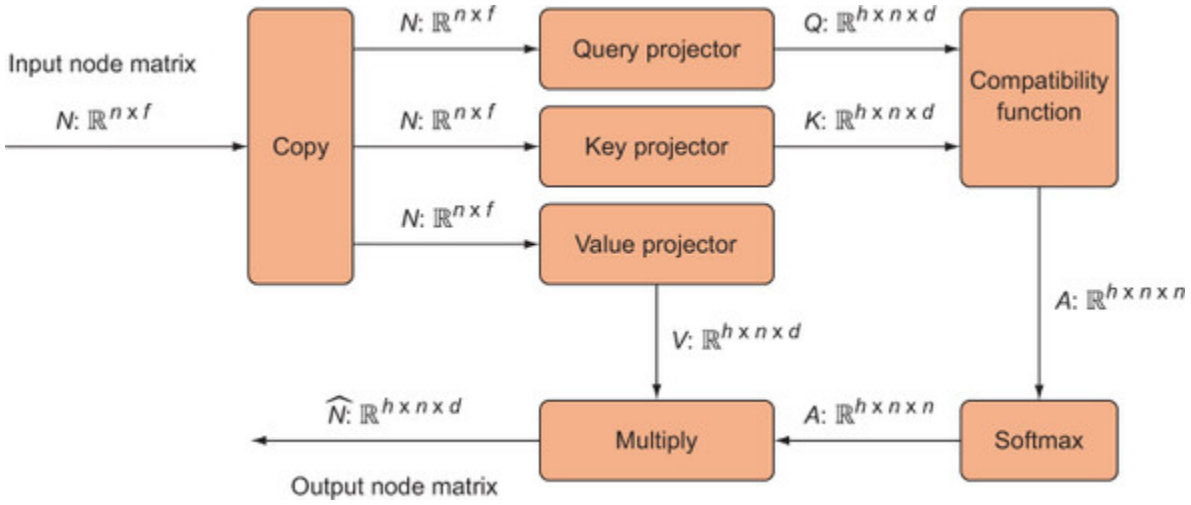
each attention head, each node can more strongly focus on a smaller subset of other nodes rather than having to spread its attention more thinly. Thus, multi-head attention can give us a better idea of which nodes are strongly related.

Figure 10.18. Multi-head dot product attention (MHDPA). Rather than use a single attention matrix, we can have multiple attention matrices called heads that can independently attend to different aspects of an input. The only difference is adding a new head dimension to the query, key and value tensors.



With multi-head attention, the utility of Einsum becomes even more obvious as we will be operating on 4-tensors of dimension *batch* × *head* × *number of nodes* × *features*. Multi-head attention will not be particularly useful for MNIST because the input space is already small and sparse enough that a single attention head has enough bandwidth and interpretability. Hence, this is a good time to introduce our reinforcement learning task for this chapter. Because the relational module is the most computationally expensive model we've implemented in this book so far, we want to use a simple environment that still demonstrates the power of relational reasoning and interpretability in reinforcement learning.

We will be coming full circle and returning to a Gridworld environment that we first encountered in chapter 3. But the Gridworld environment we will be using in this chapter is much more sophisticated. We'll be using the MiniGrid library found on GitHub at https://github.com/maximecb/gym-minigrid; it is implemented as an OpenAI Gym environment. It includes a